

Proposals should be concise: 2–3 pages recommended, 5 pages maximum. The proposal should use 11pt font or larger, 1-inch margins or larger, and single line spacing or larger, regardless of preparation format. All workshops and tutorials are half-day, in-person events. Guidance text in this template disappears automatically when `\setbool{submissionready}{true}` is activated.

From Single-Agent to Multi-Agent Physical AI

Workshop

Kexin will handle this <https://example.com>

Organizers

Name	Affiliation	Email
Ruolin Li	University of Southern California	ruolinl@usc.edu
Chen Tang	University of California, Los Angeles	ctangac@ucla.edu
Xudong Wang	Physical Intelligence	[email]
Jiachen Li	Georgia Institute of Technology	[email]
Kexin Wang	University of Southern California	[email]
Oier Mees	Microsoft, Zurich / ETH Zurich	[email] not confirmed yet
Shuijing Liu	xxx	[email]

Organizing team diversity. Geography: five US, one Europe (Zurich). Sector: four academia, two industry (Physical Intelligence, Microsoft). Career stage: a mix of junior faculty, postdocs, and doctoral researchers, alongside senior industry scientists. Gender: three women, four men.

Description

Provide a short description of the proposed event, including its motivation, relevance to the robot learning community, and intended audience.

By *Physical AI* we mean embodied agents acting in the physical world—robots, vehicles, drones, and other systems that sense, decide, and act under real-world constraints. Unlike purely digital agents, physical AI is fundamentally shaped by embodiment: physical interaction, safety, uncertainty, and deployment at scale make the underlying questions concrete and consequential.

The current scaling frontier in physical AI is driven by large datasets, foundation models, and increasingly general single-agent policies. In most of these approaches, other agents are treated as part of the environment—to be perceived, predicted, and reacted to. In contrast, the long-standing field of multi-agent robotics starts from a different premise: agents’ decisions are coupled, and the behavior of the overall system emerges from these interactions. Understanding and shaping such interactions may become essential as physical AI scales.

This workshop asks a simple but fundamental question: ****Can the multi-agent problem be deferred until single-agent capability matures, or must it shape the design of physical AI from the beginning?****

Robots are increasingly deployed not as isolated systems but as populations. Autonomous vehicles (AVs) are perhaps the clearest near-term example: robotaxi fleets already operate at city scale, and public incidents have shown vehicles clustering, blocking streets, or failing to negotiate with one another. Warehouse fleets, aerial delivery systems, and collaborative manipulation platforms are following similar trajectories. As deployment scales, the assumption that interactions are negligible becomes harder to sustain. Phenomena often ignored in single-agent settings—congestion, deadlock, oscillation, coordination failures, and social dilemmas—can directly determine system performance.

The difficulty of these challenges depends strongly on the deployment setting, and understanding this spectrum is itself a central goal of the workshop. Some populations are centrally controlled toward a shared objective—for example, a warehouse fleet operated by a single company or a cloud-hosted policy controlling many robots. In such settings, the key question is whether the population should be treated as a single high-dimensional controller or as a genuine multi-agent system. Other settings are decentralized, partially observed, and composed of agents with different owners or competing objectives, making coordination substantially more difficult.

This workshop provides a venue to think through these questions collectively: Where are single-agent methods sufficient? When do population-scale effects become unavoidable? What theories, algorithms, benchmarks, and system designs are needed to take multi-agent physical AI seriously? We bring together researchers and practitioners from academia and industry because the answers are likely to differ across deployment scenarios, and meaningful progress requires comparing experiences across domains. The workshop is aimed at researchers in robot learning, autonomous driving, warehouse and logistics robotics, aerial systems, collaborative manipulation, and multi-agent learning. Our goal is that every attendee leaves not only with new ideas, but with a clearer shared map of the challenges and opportunities at the intersection of multi-agent systems and physical AI.

Core Challenges / Research Questions

Explicitly list the specific open problems or research challenges the event aims to address. Broad survey-style themes are discouraged.

Rather than survey multi-agent robotics, the workshop targets open problems specific to populations of learned, embodied agents in the physical world:

- **Does single-robot scaling deliver multi-agent competence?** The field’s bet is that scaling data and generalist policies yields broad capability. We ask whether coordinated population behavior emerges from that recipe, or whether multi-agent is a distinct axis that scaling alone does not address.
- **Centralized or decentralized at deployment scale?** When is a robot population genuinely a multi-agent system, and when is it better treated as a single high-dimensional controller, possibly one cloud-hosted model? What decides the regime: communication and latency, a common versus misaligned objective, single versus multiple owners?
- **Coordination under misaligned objectives in shared physical space.** When fleets owned by different operators share a road, a warehouse floor, or an airspace, each optimizing its own goal, system performance can fall well below what coordination would allow. What can incentive design, mechanism design, or light-touch control achieve, and what is the realistic cost of not coordinating?
- **Simulation, benchmarks, and sim-to-real for populations.** Single-robot metrics do not capture fleet-level safety, throughput, or emergent physical effects such as congestion, deadlock, and contact. What simulators, benchmarks, and transfer methods would let the community measure population-scale progress and move it onto hardware?

Invited Speakers

For each speaker, use exactly one of the following state macros: \Requested (invited, no reply yet), \Tentative (positive signal, not fully confirmed), \Online (confirmed, but online only), \InPerson (confirmed and attending in person), \Committed (confirmed in person and committed to staying for the full workshop duration). For Workshops, the speaker list should remain small and focused, given that all events are half-day. Each speaker should be clearly motivated in terms of how they contribute to the stated challenges or instructional goals.

Sergey Levine

COMMITTED

UC Berkeley

[email]

Anchors the single-robot side: the case that scaling generalist policies is where the leverage in robot learning currently is. His perspective keeps the central question honest, since deferring multi-agent is a defensible position, and speaks directly to whether single-robot scaling delivers multi-agent competence.

Peter Stone

COMMITTED

UT Austin / Sony AI

[email]

Brings the foundations of multi-agent systems, including ad hoc teamwork: coordinating with agents without shared information or prior agreement. Speaks to the centralized-versus-decentralized and misaligned-objective questions, and his role at Sony AI adds a deployment perspective.

Dorsa Sadigh

not confirmed yet

Stanford University

[email]

Brings interaction-aware robotics and coordination with agents whose objectives differ, speaking to coordination under misaligned objectives.

Ziwei Liu

COMMITTED

Nanyang Technological University, Singapore

[email]

Brings large-scale visual foundation models, generative world models, and embodied AI, speaking to whether the single-robot scaling recipe and learned world models extend to population settings, and to simulation for population-scale behavior.

Speaker diversity. Gender: one woman, three men. Geography: three US, one Asia (Singapore). Sector: academia, with industry perspective through Sony AI and Physical Intelligence ties.

Beyond the speakers, we will separately invite three to four panelists for the panel discussion, chosen to broaden the range of voices: early-career researchers, industry practitioners, and members of underrepresented groups. This keeps the talk lineup small, as the Call for Workshops asks, while widening participation and perspective.

to be confirmed

Additional panelists. Boyi Li (NVIDIA),

Panelist diversity. Gender: one woman, three men. Geography: three US, one Asia (Singapore). Sector: academia, with industry perspective through Sony AI and Physical Intelligence ties.

Proposed Format and Interactivity

Describe the format of the event and how it actively engages participants. Passive sequences of talks are discouraged. Include interactive elements such as breakout discussions, collaborative problem-solving, live demos, fishbowl discussions, or hands-on components, as appropriate. Ensure the format is suitable for a half-day event.

The half day pairs framing talks and contributed presentations with a moderated panel as its interactive core. We chose a panel of separately invited panelists driven by live audience questions over breakout-style activities, so the session stays engaging while remaining light to run: it needs only a moderator and microphones. An indicative schedule (about 4 hours):

- **Opening and framing (10 min).** The organizers pose the central question and the four challenges.
- **Invited position statements (120 min).** Four talks, about 30 minutes each, from the invited speakers, framed as positions across the single-to-multi spectrum rather than result surveys.
- **Panel discussion and Q&A (40 min).** A moderated panel of separately invited panelists works through the central question and the audience-submitted questions, anchored in concrete settings (AV fleets, warehouse fleets, drone delivery). Attendees also submit discussion questions through a live tool (e.g., Slido), which seed the panel.
- **Contributed paper presentations (30 min).** Short oral talks, about 10 minutes each, from the top-rated accepted papers running for award.
- **Poster and coffee session (30 min).** The remaining accepted papers appear as posters.
- **Best paper award and closing (15 min).** Presentation of the sponsor-backed best paper award, followed by a short synthesis of the open challenges that seeds the post-workshop artifact.

Contributed Papers Plan

Describe plans for soliciting, reviewing, and selecting contributed papers.

We will solicit short papers (up to 4 pages plus references) on new, preliminary, or in-progress work in multi-agent and population-scale robot learning. Submissions will be handled on OpenReview and reviewed double-blind by a small program committee drawn from the organizers’ networks and broadened for diversity. Consistent with CoRL policy, we will not accept work already accepted to the main CoRL 2026 conference; the workshop is a space for early ideas. The top-rated accepted papers are invited to give short oral presentations; the remainder appear as posters during the coffee session. A confirmed sponsor backs a best paper award, presented during the closing, which we use to draw strong submissions from students and first-time contributors; the same sponsorship funds student travel (see Participation and Diversity).

Plans to Encourage Participation and Diversity

Describe concrete plans to encourage participation beyond the invited speaker list. Also describe specific efforts to promote diversity across career stage, gender, geography, institution, and background.

Because the four invited speakers are senior, we deliberately build diversity into the other roles, which the interactive format gives us room to do. A confirmed sponsor has committed funding earmarked for broadening participation, so these are backed by real resources rather than intentions:

- **Funded student travel grants**, supported by our sponsor, will prioritize student and first-time presenters who would otherwise be unable to attend, with attention to gender, race and ethnicity, geography, and institution.
- **Separately invited panelists** for the panel will include Ph.D. students, postdocs, and industry practitioners from autonomous vehicles, warehouse robotics, and aerial delivery, broadening perspective beyond the senior speaker lineup.

- **Contributed papers** will be selected with attention to a diverse author pool, with lightning slots reserved for student-led work and a sponsor-backed award to encourage strong first-time submissions.
- **Outreach** will extend beyond our networks through relevant mailing lists and affinity groups, and through an open call on the workshop website.

Logistics and Resources

Estimated audience size: 80 to 120 attendees at peak, with 40 to 60 sustained through the interactive sessions.

Physical requirements:

Please provide your estimate of audience size (virtual and in-person) and other physical resources needed for the workshop, e.g., poster boards, etc.

A single room with a panel area (table and seating for panelists) at the front, a projector and screen, roving microphones for audience questions, and 10 to 15 poster boards with power nearby. No breakout rooms are required.

Special requirements:

Are there any special requirements for the workshop or tutorial?

None beyond the seating configuration above. The event is in person. Funding for student travel grants and the contributed-paper award is already secured through a confirmed sponsor.

Organizer Background and Experience

The organizing team combines complementary expertise across academia and industry: Ruolin Li (control and incentive design for populations of autonomous agents, mixed-autonomy traffic), Chen Tang (autonomous driving, interaction, and behavior prediction), Kexin Wang ([research area]), Jiachen Li (multi-agent interaction and trajectory prediction), Xudong Wang (robot learning, Physical Intelligence), and Oier Mees (robot foundation models and generalist policies, Microsoft and ETH Zurich).

The team brings substantial workshop-organizing experience. Oier Mees has co-organized workshops on language and robot learning and on scaling robot learning at venues including CoRL, RSS, and ICRA. [Add the remaining organizers' prior workshops, tutorials, and program-committee or area-chair roles here, with venues and years.]

Commitments

- ✓ At least one organizer will attend CoRL 2026 in person.

Confirm that at least one organizer will attend CoRL 2026 in person, and list which organizer(s) will attend.

Attending in person: [list organizer(s), e.g., Ruolin Li, Chen Tang].

- ✓ Organizers will attend the CoRL 2026 workshop organizers meeting.

Confirm commitment to attend the virtual CoRL 2026 workshop organizers meeting following acceptance.

- ✓ Organizers will submit a post-workshop artifact.

Confirm commitment to submit a post-workshop artifact, such as a whitepaper, community report, or position paper, following the Call for Workshops requirements.

We will produce a community position paper, “When does multi-agent matter for physical AI?”, synthesizing the panel discussion and audience-submitted questions into a shortlist of open challenges and candidate directions.

✓ Willingness to merge with related proposals if requested.

Indicate whether you would be open to merging with another high-quality proposal in the case of overlapping topics.